Vibration Sensor

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***Abstract*— This paper presents the design and development of a low-cost, analog-based vibration sensor for monitoring mechanical systems, such as motors and industrial machinery, within a frequency range of 100 Hz to 500 Hz. The proposed sensor employs a piezoelectric transducer to convert mechanical vibrations into an electrical signal, where the voltage variation of the signal is measured to determine the vibration intensity.**

I. Introduction

The project is conducted to develop a vibration sensor that primarily measures vibration levels in mechanical systems, such as industrial machinery and structural components. Initial research was conducted on vibration sensor principles commonly used in the industry. Several principles were explored, including electrodynamic, capacitive, and piezoelectric methods. However, the capacitive method, while initially considered, produced capacitance changes that were too small to measure accurately with the available analog setup, leading to unreliable results. Non-contact methods, such as optical fiber sensors and microphone-based sensors, were also evaluated, as they are often used in commercial applications. However, optical fiber sensors are expensive and beyond our scope, and microphone-based sensors are highly sensitive to ambient noise, which can interfere with accurate measurements in industrial environments. Due to these constraints, the best choice was a piezoelectric method to measure vibrations effectively, leveraging its high sensitivity and ability to generate measurable electrical signals.

As Electrical and Electronics Engineering undergraduates, the plan was to generate a voltage signal that varies with the vibration intensity. To implement this, a piezoelectric vibration sensor was designed, utilizing a piezoelectric crystal and seismic mass to convert mechanical vibrations into an electrical charge. Since the generated signal was small, operational amplifiers were employed to amplify the signal. The specialty of this project is that no microcontroller or other complex technologies were used to process the signal. Basic analog components such as operational amplifiers for signal amplification were used to implement the circuit. The sensor can measure vibrations in the frequency range of 100 Hz to 500 Hz, with potential modifications to enhance sensitivity or accuracy for specific applications.

The intended application of this sensor is to be attached to mechanical systems to measure vibration levels with high accuracy using the piezoelectric method. It can be highly useful for industrial factories, railway crossing automation, structural health monitoring, and seismic activity detection. As the next step, an alarm system or automated control unit can be designed with this sensor module to trigger alerts or maintenance actions when excessive vibrations are detected.

Initial specifications

A. Measurement Quality

This sensor is applicable to any mechanical system that the piezoelectric element contacts, but its performance depends on the material of the system, as the frequency response generated by the material alters the calibration curve.

B. Range

The sensor is designed to measure amplitude(vibration) of 0.1mm – 2.5mm in the range of 100Hz to 500Hz

C. Other specifications

- Operating Voltage: 1 V – 25V
- Operating Temperature: -20° - 60°
- Error ±0.05mm

II. Method

A. Principle of operation

Initially, the piezoelectric sensor, consisting of a piezoelectric crystal and a seismic mass, is securely attached to the vibrating mechanical system, such as a motor or structural component. When the system vibrates within the frequency range of 100 Hz to 500 Hz, the mechanical stress induced by the vibrations causes the piezoelectric crystal to generate an electrical charge via the piezoelectric effect. This charge is converted into an analog voltage signal proportional to the vibration intensity[1]. However, the voltage generated by the piezoelectric sensor is typically small, ranging from a few millivolts, depending on the vibration amplitude. Therefore, operational amplifiers are employed to amplify the signal to a suitable range for further processing.

B. Circuit diagrams

This sensor has two main parts.

- Sensor module
- Amplification module
- Output Module

These circuits contain Piezoelectric sensor, Amplifier which is powered by two 9V power sources.

C. *Amplification circuit*

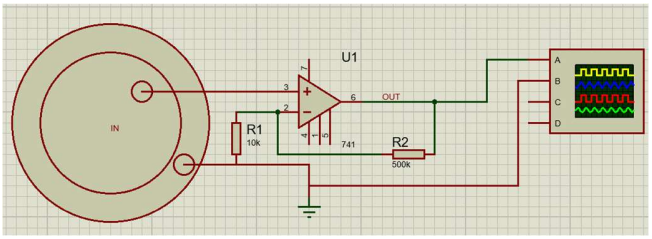


Fig. 2. Amplification circuit

This Amplification circuit contains a LM741 op-amp, 500kΩ feedback resistor and 10kΩ input resistor.

D. *Sensor module*

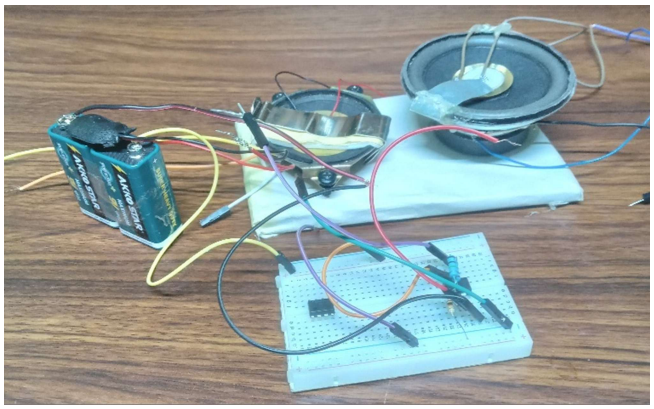


Fig. 2. Calibration setup

The speaker is powered by a frequency generator, and the piezoelectric sensor gets the input from the speaker's vibration. The signal is then sent to an amplifier circuit and displayed on an oscilloscope.

III. RESULTS

A. *Observations*

TABLE I. VARIATION OF THE OUTPUT VOLTAGE WITH AMPLITUTE OF THE VIBRATION

Amplitude(mm)	Output Voltage(mV)		
	100Hz	200Hz	300Hz
0.1	240	200	200
0.2	400	200	400
0.3	500	400	500
0.4	800	500	650
0.5	1000	500	680
0.6	1180	600	850
0.7	1280	760	900
0.8	1600	800	1050
0.9	1880	850	1200
1.0	2000	900	1300
1.1	2200	950	1450
1.2	2400	1000	1600
1.3	2600	1150	1820
1.4	2800	1200	1950
1.5	2950	1200	2320
1.6	3100	1250	2610
1.7	3250	1300	3000
1.8	3450	1350	3300
1.9	3550	1350	4000
2.0	3800	1350	4200
2.1	4000	1350	5500
2.2	4100	1400	4900
2.3	4100	1500	4600
2.4	4250	1550	4700
2.5	4300	1600	5150

B. *Input and output characteristics graph*

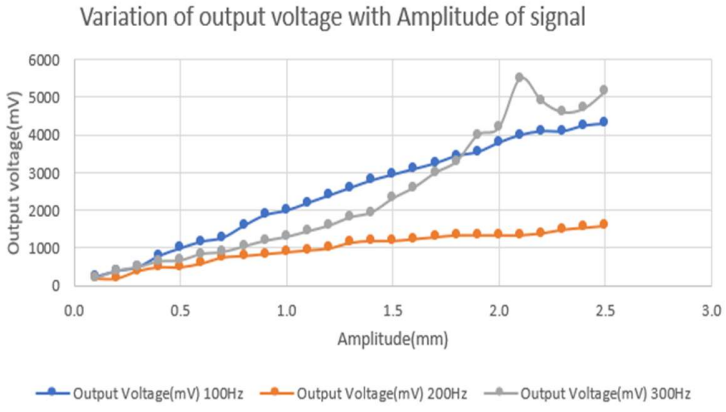


Fig. 3. Variation of output voltage with amplitude of signal

IV. Discussion

A. Method Selection

According to the proposal, a capacitive vibration sensor was initially explored, measuring vibrations via capacitance changes between two plates. However, the required 0.1 mm plate separation and 1 pF series capacitance posed fabrication challenges, leading to inconsistent results. Consequently, we shifted to a piezoelectric sensor, which generates voltage from mechanical stress, offering a more reliable and practical approach for vibration detection.

B. Finding a Way to Measure the Vibration

To implement the piezoelectric method, we researched journals, articles, and online resources on piezoelectric sensors for vibration measurement. The chosen approach involved mounting a piezoelectric element on a vibrating surface to convert mechanical oscillations into electrical signals. These signals were then processed to correlate with vibration amplitude and frequency, ensuring effective detection within the 100 Hz to 500 Hz range.

C. Designing the Sensor Circuit

To reduce costs and avoid the complexity of microcontrollers, this basic sensor module excludes an Arduino or any microcontroller. However, as a future modification, integrating an Arduino could convert the sensor into a seismometer capable of reading seismic earth vibrations, enhancing its functionality and specifications[2]. Working with basic components like the piezoelectric sensor and an LM358 op-amp for amplification was challenging but cost-effective.

D. Accessibility to the Components

An oscilloscope and signal generators are crucial for calibrating this sensor module. However, the oscilloscope was only available during lab hours, limiting our access and making calibration challenging as beginners. Fortunately, other components, such as the piezoelectric elements, op amp LM741 IC, and Resistors, were easily accessible locally, allowing us to proceed with circuit assembly and testing without significant delays in sourcing materials.

E. Troubleshooting

As mentioned, troubleshooting was difficult without an oscilloscope and signal generator. To overcome this, we used a speaker to play the desired frequency from a Bluetooth-connected phone as a signal source, simulating vibration inputs for testing. The Arduino serial plotter was employed to analyze the output signal from the piezoelectric sensor, allowing us to monitor its behavior. Additionally, a multimeter was used to inspect components and connections individually, helping us identify and fix errors in the circuit to ensure proper operation.

F. Strengths

- Simplicity– The sensor uses basic electronic components and does not require a microcontroller, reducing both cost and system complexity.
- Reliability– The piezoelectric response ensures consistent performance over time with minimal signal drift.
- Compactness – The lightweight design allows for easy installation on various surfaces without adding significant bulk.
- Ease of Installation – No specialized knowledge of electronics or sensors is needed for setup, making it accessible for both industrial and home use.

G. Weaknesses

- Limited Frequency Range – The sensor is only effective for medium-frequency vibrations (100 Hz to 500 Hz), making it unsuitable for ultra-low or high-frequency applications.
- Manual Calibration Required – After installation on a vibrating surface, the sensor must be recalibrated each time to ensure accurate measurements, adding to maintenance efforts.
- Resolution Limitations – The current design provides only basic vibration detection without fine granularity in measurement. Future enhancements with signal processing could improve resolution.

V. Conclusion

Our piezoelectric vibration sensor system demonstrated reliable performance in measuring mechanical vibrations with considerable accuracy, proving its effectiveness for industrial and structural monitoring applications. By leveraging analog signal conditioning techniques, we successfully transformed small piezoelectric signals into robust, measurable outputs without relying on complex digital processing. This project highlights the practicality of piezoelectric sensing in real-world scenarios, offering a cost-effective and efficient solution for vibration detection in machinery, infrastructure, and seismic monitoring.

Further research could explore energy harvesting from vibrations to create self-powered sensor networks[3]. Our work not only contributes to the advancement of vibration sensing technology but also serves as a foundation for developing smarter, more responsive industrial monitoring systems. Moving forward, expanding the sensor's frequency range and refining its sensitivity could unlock new applications in predictive maintenance and automated safety systems, further pushing the boundaries of vibration sensing innovation.

References

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